Kelvin Jusitne Obi
VUG/SEN/24/12727

Part 2: Normalization from 1NF to 3NF

VUA-SEN307 Week 4 Practical

The starting sheet

The lists in each row line up by position, so each course is paired with its title and lecturer in the same order. This is the assumption used throughout.

Student	Course	Title	Lecturer	Lecturer's dept
Ada Obi	SEN307	Database Systems	Dr A. Okafor	Software Eng.
Ada Obi	SEN301	Software Architecture	Dr M. Bello	Software Eng.
John Musa	SEN307	Database Systems	Dr A. Okafor	Software Eng.
John Musa	COS301	Operating Systems	Mrs F. Eze	Computer Sci.
Grace Okon	SEN301	Software Architecture	Dr M. Bello	Software Eng.

Step 1: First Normal Form (1NF)

Each student-course pair now has its own row, so every cell holds one value.

matric_no	student_name	student_dept	course_code	course_title	lecturer	lecturer_dept	session
20231234	Ada Obi	Software Engineering	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231234	Ada Obi	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027

matric_no	student_name	student_dept	course_code	course_title	lecturer	lecturer_dept	session
20231235	John Musa	Computer Science	COS301	Operating Systems	Mrs F. Eze	Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

Primary key: matric_no + course_code. Ada appears twice, so matric_no alone is not unique. SEN307 appears twice, so course_code alone is not unique.

What I fixed: I split the comma-separated lists so every cell holds one value, giving each student-course pair its own row, and chose matric_no + course_code as the composite key.

Step 2: Second Normal Form (2NF)

For each column, I asked: do I need matric_no, course_code, or both to know this?

Column	Needs	Verdict	Moves to
student_name, student_dept	matric_no only	Partial dependency	Student
course_title	course_code only	Partial dependency	Course
lecturer, lecturer_dept	course_code and session, not the student	Partial dependency	Teaching
session	the enrolment (both)	Full dependency	Enrolment

Student

matric_no (PK)	student_name	student_dept
20231234	Ada Obi	Software Engineering
20231235	John Musa	Computer Science
20231236	Grace Okon	Software Engineering

Course

course_code (PK)	course_title
SEN307	Database Systems
SEN301	Software Architecture
COS301	Operating Systems

Enrolment

matric_no (FK)	course_code (FK)	session
20231234	SEN307	2026/2027
20231234	SEN301	2026/2027
20231235	SEN307	2026/2027
20231235	COS301	2026/2027
20231236	SEN301	2026/2027

Teaching (key: course_code + session)

course_code	session	lecturer	lecturer_dept
SEN307	2026/2027	Dr A. Okafor	Software Eng.
SEN301	2026/2027	Dr M. Bello	Software Eng.
COS301	2026/2027	Mrs F. Eze	Computer Sci.

Why a separate Teaching table: who teaches a course depends on the course and the session, not on any student. It cannot stay in Enrolment. It cannot go in Course either, because a course can have different lecturers in different sessions.

What I fixed: I moved student_name and student_dept to Student, course_title to Course, and the lecturer columns to their own table, because each depended on only part of the key.

Step 3: Third Normal Form (3NF)

For each column I asked: does this describe the key, or does it describe another non-key column?

Where	Column	Problem	Fix
Student	student_dept	A department's details belong to the department, not the student. If the department is renamed, every student row must be edited.	Department table, linked by dept_id
Teaching	lecturer_dept	It describes the lecturer, not the course session.	Lecturer table, which holds dept_id
Teaching	lecturer (name)	A lecturer is their own thing, with a department attached.	Lecturer table, referenced by lecturer_id

Department

dept_id (PK)	dept_name
1	Software Engineering

dept_id (PK)	dept_name
2	Computer Science

Student

student_id (PK)	matric_no	full_name	dept_id (FK)
1	20231234	Ada Obi	1
2	20231235	John Musa	2
3	20231236	Grace Okon	1

Lecturer

lecturer_id (PK)	full_name	dept_id (FK)
1	Dr A. Okafor	1
2	Dr M. Bello	1
3	Mrs F. Eze	2

Course

course_code (PK)	course_title
SEN307	Database Systems
SEN301	Software Architecture
COS301	Operating Systems

Enrolment

student_id (FK)	course_code (FK)	session
1	SEN307	2026/2027
1	SEN301	2026/2027
2	SEN307	2026/2027
2	COS301	2026/2027
3	SEN301	2026/2027

Course_Assignment (records which lecturer teaches which course in which session)

course_code (FK)	lecturer_id (FK)	session
SEN307	1	2026/2027
SEN301	2	2026/2027

course_code (FK)	lecturer_id (FK)	session
COS301	3	2026/2027

What I fixed: I replaced the department text with a dept_id foreign key and split lecturers into their own table, so each department name is written once and a rename means updating one row.

Step 4: Final schema

- Department (dept_id PK, dept_name)
- Student (student_id PK, matric_no, full_name, dept_id FK → Department)
- Lecturer (lecturer_id PK, full_name, dept_id FK → Department)
- Course (course_code PK, course_title)
- Enrolment (student_id FK → Student, course_code FK → Course, session), PK = (student_id, course_code, session)
- Course_Assignment (course_code FK → Course, lecturer_id FK → Lecturer, session), PK = (course_code, lecturer_id, session)